

# LSI\_UNED at #SMM4H-HeaRD 2026: Grid-Based Biomedical Named Entity Recognition Across Languages and Entity Types

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## Abstract

This paper describes the participation of the LSI\_UNED team in the first sub-task of MultiClinAI at the #SMM4H-HeaRD 2026 Workshop, which focuses on multilingual clinical named entity recognition in seven languages. The task requires identifying mentions of diseases, procedures, and symptoms in clinical case reports. We propose a set of systems based on the W<sup>2</sup>NER architecture, with a separate model trained for each language and entity type. For Spanish, we use a RoBERTa-based model with data augmentation from additional NER resources, while English and Italian systems are based on different biomedical BERT variants. Results show consistent performance across languages, with the best overall results obtained for Spanish. Data augmentation improves recall and F1, while English and Italian models achieve competitive but slightly lower scores. Symptom recognition remains the most challenging entity type across all languages.

## 1 Introduction

The Social Media Mining for Health Applications and Health Real-World Data (#SMM4H-HeaRD) 2026 Workshop (Lopez-Garcia et al., 2026) encompasses eight shared tasks that focus on natural language processing (NLP), machine learning, and artificial intelligence using health-related data sources. Task 8, called MultiClinAI (Gallego-Donoso et al., 2026), addresses the growing need for multilingual corpora by evaluating systems that recognize medical concepts across seven languages. Its first sub-task, MultiClinNER, focuses on identifying diseases, procedures, and symptoms in clinical case reports written in Spanish, English, Dutch, Italian, Romanian, Swedish, and Czech. Participants must detect entity mentions and return their exact character offsets.

This paper describes the participation of our team, LSI\_UNED, in MultiClinNER for the Span-

ish, English, and Italian languages. Because of time constraints, we could not apply the system to all seven languages, so we selected the three with which we are most familiar. We used the W<sup>2</sup>NER (Li et al., 2022) architecture, adapted for the task. The fine-tuned systems were configured for a single-label and monolingual task. For the Spanish texts, we used a RoBERTa-based pre-trained model and also augmented the training set with additional NER resources. For English and Italian, two different BERT-based models were fine-tuned.

## 2 Related work

Named Entity Recognition (NER) is a widely addressed NLP task that converts unstructured text into structured information by detecting and classifying entities. In the medical domain, it is essential for identifying healthcare-related concepts and supports many downstream applications (Goyal and Singh, 2025; Ahmad et al., 2023).

In Spanish, several benchmarks address clinical entity extraction, including DisTEMIST (Miranda-Escalada et al., 2022) for diseases, MedProcNER (Lima-López et al., 2023) for procedures, SympTEMIST (Lima-López et al., 2023) for symptoms, and a sub-task of Cantemist (Miranda-Escalada et al., 2020) for tumor morphology. In English, GENIA (Kim et al., 2003) focuses on biological terms, MedMentions (Mohan and Li, 2019) on a broad range of biomedical entities, and CHEMDNER (Krallinger et al., 2015) on chemical and drug-related mentions. Italian has fewer resources, such as CLARO (Paolo et al., 2023), which covers 25 clinical entity types. Furthermore, as previously mentioned, the MultiClinAI shared task covers a total of seven languages, making it a valuable setting in the multilingual biomedical field. Other multilingual clinical NER corpora include MANTRA (Kors et al., 2015), which aligns biomedical concepts across English, Spanish, French, German,

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| English      |               |               | Spanish      |               |               | Italian      |               |               |
|--------------|---------------|---------------|--------------|---------------|---------------|--------------|---------------|---------------|
| Label        | Total         | Unique        | Label        | Total         | Unique        | Label        | Total         | Unique        |
| DISEASE      | 25,118        | 11,783        | DISEASE      | 26,296        | 13,097        | DISEASE      | 26,159        | 12,584        |
| PROCEDURE    | 26,733        | 9,746         | PROCEDURE    | 28,137        | 10,642        | PROCEDURE    | 27,394        | 9,936         |
| SYMPTOM      | 27,465        | 13,878        | SYMPTOM      | 29,074        | 14,798        | SYMPTOM      | 27,929        | 14,156        |
| <b>Total</b> | <b>79,316</b> | <b>35,407</b> | <b>Total</b> | <b>83,507</b> | <b>38,537</b> | <b>Total</b> | <b>81,482</b> | <b>36,676</b> |

Table 1: Statistics of the MultiClinNER corpus across the three evaluated languages (English, Spanish, and Italian), including the number of total and unique entity mentions for the three labels: DISEASE, PROCEDURE, and SYMPTOM.

and Dutch, and the E3C corpus (Magnini et al., 2020), a collection of clinical case reports in English, French, Italian, Spanish, and Basque.

Several architectures have been proposed for NER both in the general and biomedical domains. The most common approach is token-level sequence labeling, where each token is assigned a tag using schemes such as IOB2 or BILOU, with models based on CRF, BiLSTM-CRF, or Transformers (Zhang et al., 2018; Pooja and Jagadeesh, 2024). Another approach is span-based NER, which predicts entity boundaries directly instead of labeling tokens one by one. In this group, grid-based models such as W<sup>2</sup>NER (Li et al., 2022) represent word-pair relations in a two-dimensional grid, which helps to detect nested or complex entities. More recent approaches include generative NER, which reformulates entity extraction as a text generation task (Yuan et al., 2022; Lv et al., 2025).

### 3 Methodology

#### 3.1 Dataset

The MultiClinNER corpus (Lima López et al., 2026) contains 1,258 documents per language. For each language, it is annotated with three medical concepts: diseases, procedures, and symptoms. Table 1 presents the number of total and unique entities per label across the three languages for which our team submitted results. As can be observed, there are more than 25,000 mentions per entity type in each language. Moreover, the label with the fewest unique mentions is PROCEDURE, whereas SYMPTOM is the label with the highest number of both total and unique mentions, possibly indicating that it is the most challenging entity type.

#### 3.2 System description

##### 3.2.1 Grid-based NER method

The NER architecture used in this work is based on W<sup>2</sup>NER, proposed in (Li et al., 2022). Unlike traditional sequential tagging approaches, W<sup>2</sup>NER

reformulates NER as a word-pair relation classification task using a two-dimensional grid. Each cell in the grid represents a pair of words, and the model predicts relations such as Next-Neighboring-Word (NNW), which links adjacent words within the same entity, and Tail-Head-Word (THW), which connects the last and first words of an entity while assigning its label. The architecture combines BERT embeddings, a BiLSTM for contextual encoding, and 2D convolution layers to capture interactions between word pairs. Entities are then reconstructed by linking words through the predicted relations. The original work reports state-of-the-art results on 14 NER benchmark datasets.

##### 3.2.2 Pre-trained model selection and training

Different pre-trained models were used for each language. All of them are monolingual and pre-trained on biomedical texts. They are described below.

- English: For the English texts, two pre-trained models were used. The first was biobert-v1.1 (110M parameters) (Lee et al., 2020), which was pre-trained on large biomedical corpora, including PubMed abstracts and PubMed Central full-text articles. The second was Bio\_ClinicalBERT (110M parameters) (Alsentzer et al., 2019), which was initialized from the former model and further pre-trained on clinical notes from the MIMIC-III database (Johnson et al., 2016).
- Spanish: The pre-trained model used for the Spanish texts was roberta-base-biomedical-clinical-es (125M parameters) (Carrino et al., 2021). It is based on the Spanish RoBERTa-base model and was further pre-trained on multiple clinical Spanish corpora collected from public sources and a dataset containing more than 278,000 documents.
- Italian: The pre-trained models used for the Italian texts were bioBIT (100M parameters)

(Buonocore et al., 2023), trained on Italian medical texts, and MedPsyNIT (100M parameters) (Crema et al., 2023), developed for medical and psychological text processing in Italian.

Due to time constraints, we were unable to perform hyperparameter optimization. The parameters were set to a batch size of 8, a learning rate of  $1e^{-3}$ , and a maximum sequence length of 512 tokens per chunk, with chunks segmented at the sentence level. Nevertheless, we conducted 5-fold cross-validation to select the optimal number of training epochs. With a maximum of 35 epochs, the best micro F1 score was consistently reached at approximately 20 epochs on average across folds. Experiments were conducted on a system equipped with an NVIDIA A30 GPU featuring 24 GB HBM2 ECC memory. Fine-tuning each model took 8 to 10 hours.

### 3.2.3 Data augmentation

For the Spanish corpus, we decided to augment the training set because several biomedical NER resources in Spanish have been published in recent years. Data augmentation was not performed for the English and Italian corpora, as we did not find corpora annotated with the same entity types required for the task. We used the corpora from the DisTEMIST, MedProcNER, and SympTEMIST tasks (described in Section 2) to augment the training data for diseases, procedures, and symptoms, respectively. Furthermore, we employed the CARMEN-I resource (Farre Maduell et al., 2024), which also includes annotations for the three entity types. The CARMEN-I corpus contains documents in Spanish, Catalan, and bilingual texts. Therefore, we selected only the texts written in Spanish and incorporated their annotations to further augmenting the training set. After the evaluation phase, we noticed that a subset of the added documents overlapped with the original texts, resulting in only 33–35% of the added documents being unseen. Nonetheless, the results in Section 4 show that data augmentation still improves the performance of the system.

## 4 Experiments and results

The evaluation of the task considers six metrics in total. The *strict* metrics (strict\_precision, strict\_recall, strict\_f1) measure exact match performance, where a prediction is only correct if it perfectly matches the gold annotation in en-

tity type and span boundaries. The *char* metrics (char\_precision, char\_recall, char\_f1) evaluate character-level overlap between predicted and gold spans, where precision is the proportion of the predicted span that overlaps with the gold span and recall is the proportion of the gold span covered by the prediction.

Table 2 shows the final results, reporting, for each language and label, the values obtained for the six evaluation metrics considered. Results are ranked by strict\_f1, and the numbers in parentheses indicate the ranking achieved in the shared task. For each language, two different runs were submitted. Each system is single-label, meaning that the models were fine-tuned separately for each entity type. The main reason for this was that the English and Italian texts showed slight variations across labels, so the offset alignments were not identical for each entity type. For Spanish, the training texts were correctly aligned across labels, but experiments with a multi-label system in a 5-fold cross-validation setting produced lower results than the single-label approach, so we chose the simpler and more consistent method.

For the English documents, the system that employs the model biobert-v1.1 (conf1\_biobert) achieved better results across the three labels, with an improvement of around 0.02 in strict\_f1 compared with the system fine-tuned from Bio\_ClinicalBERT (conf1\_clinical). The system conf1\_biobert achieved its best ranking in strict\_recall for the label PROCEDURE, where it placed second.

With respect to the Spanish systems, those trained with the augmented training set (conf2) reached a higher strict\_f1 than those using the original training set (conf1) across the three labels. In such cases, we can see that the strict\_recall is higher than in conf1, especially in DISEASE and SYMPTOM, suggesting that the data augmentation allows the model to identify more entities. For the label DISEASE, the system conf2 ranked first in strict\_recall, while for all three labels it achieved third place in strict\_f1.

Regarding the Italian language, the system fine-tuned on the model bioBIT (conf1\_biobit) achieved better strict\_f1 in the labels DISEASE and SYMPTOM, while the system using the model MedPsyNIT (conf1\_medpsynit) reached a better score in PROCEDURE. However, the differences in strict\_f1 scores are lower than 0.01 in all cases. The system conf1\_biobit achieved its best ranking

| English   |                 |                  |               |                   |                |             |                   |
|-----------|-----------------|------------------|---------------|-------------------|----------------|-------------|-------------------|
| Entity    | System          | strict_precision | strict_recall | strict_f1         | char_precision | char_recall | char_f1           |
| DISEASE   | conf1_biobert   | 0.7633 (7)       | 0.7914 (3)    | <b>0.7771</b> (5) | 0.8421 (8)     | 0.8523 (8)  | <b>0.8472</b> (7) |
| DISEASE   | conf1_clinical  | 0.7574 (8)       | 0.7572 (9)    | 0.7573 (8)        | 0.8395 (9)     | 0.8226 (13) | 0.8309 (12)       |
| PROCEDURE | conf1_biobert   | 0.7584 (4)       | 0.7293 (2)    | <b>0.7436</b> (3) | 0.8595 (7)     | 0.8074 (7)  | <b>0.8326</b> (6) |
| PROCEDURE | conf1_clinical  | 0.7531 (5)       | 0.6979 (9)    | 0.7244 (6)        | 0.8516 (8)     | 0.7739 (10) | 0.8109 (10)       |
| SYMPTOM   | conf1_biobert   | 0.7433 (6)       | 0.6802 (7)    | <b>0.7104</b> (6) | 0.8308 (6)     | 0.7409 (10) | <b>0.7833</b> (6) |
| SYMPTOM   | conf1_clinical  | 0.7397 (7)       | 0.6609 (10)   | 0.6981 (7)        | 0.8272 (7)     | 0.7220 (12) | 0.7710 (9)        |
| Spanish   |                 |                  |               |                   |                |             |                   |
| Entity    | System          | strict_precision | strict_recall | strict_f1         | char_precision | char_recall | char_f1           |
| DISEASE   | conf2           | 0.7852 (5)       | 0.8432 (1)    | <b>0.8132</b> (3) | 0.8495 (5)     | 0.8973 (3)  | <b>0.8728</b> (4) |
| DISEASE   | conf1           | 0.7925 (4)       | 0.8231 (5)    | 0.8075 (5)        | 0.8545 (4)     | 0.8758 (8)  | 0.8650 (7)        |
| PROCEDURE | conf2           | 0.8018 (2)       | 0.8049 (4)    | <b>0.8034</b> (3) | 0.8730 (2)     | 0.8662 (10) | <b>0.8696</b> (4) |
| PROCEDURE | conf1           | 0.8011 (3)       | 0.8004 (6)    | 0.8008 (4)        | 0.8699 (4)     | 0.8559 (11) | 0.8628 (7)        |
| SYMPTOM   | conf2           | 0.7472 (4)       | 0.7667 (4)    | <b>0.7568</b> (3) | 0.8226 (4)     | 0.8335 (5)  | <b>0.8280</b> (4) |
| SYMPTOM   | conf1           | 0.7626 (2)       | 0.7488 (5)    | 0.7556 (4)        | 0.8365 (2)     | 0.8066 (9)  | 0.8213 (5)        |
| Italian   |                 |                  |               |                   |                |             |                   |
| Entity    | System          | strict_precision | strict_recall | strict_f1         | char_precision | char_recall | char_f1           |
| DISEASE   | conf1_biobit    | 0.7058 (8)       | 0.7245 (2)    | <b>0.7150</b> (5) | 0.8202 (9)     | 0.8188 (4)  | <b>0.8195</b> (5) |
| DISEASE   | conf1_medpsynit | 0.7163 (7)       | 0.7102 (4)    | 0.7132 (6)        | 0.8250 (7)     | 0.8014 (6)  | 0.8130 (7)        |
| PROCEDURE | conf1_medpsynit | 0.7124 (5)       | 0.6320 (9)    | <b>0.6698</b> (7) | 0.8502 (3)     | 0.7436 (10) | 0.7933 (9)        |
| PROCEDURE | conf1_biobit    | 0.6888 (7)       | 0.6377 (8)    | 0.6623 (9)        | 0.8428 (6)     | 0.7636 (9)  | <b>0.8013</b> (8) |
| SYMPTOM   | conf1_biobit    | 0.6898 (5)       | 0.6167 (5)    | <b>0.6512</b> (4) | 0.8170 (4)     | 0.7076 (9)  | 0.7584 (7)        |
| SYMPTOM   | conf1_medpsynit | 0.6802 (6)       | 0.6241 (4)    | 0.6509 (6)        | 0.8119 (6)     | 0.7187 (8)  | <b>0.7624</b> (6) |

Table 2: Final results of the system for each label and language, ranked by strict\_f1. The numbers in parentheses indicate the ranking achieved for each metric with respect to the other participating teams. For each language and label, the best strict\_f1 and char\_f1 results are highlighted in bold.

in strict\_recall for the label DISEASE, where it placed second.

To conclude, the overall results seem consistent and robust across languages and labels. In general, the obtained results in the Spanish corpus were the best ones. Moreover, the last three metrics are higher than the first three in all cases, as they consider the character-overlap of the predictions instead of counting non-exact matchings as errors. Furthermore, it can be observed that in most cases, the models make more mistakes and obtain lower results when detecting entities corresponding to the label SYMPTOM. This aligns with the hypothesis made in Subsection 3.1, referring to the high number of unique mentions presented by this label.

## 5 Conclusions

This work presents the participation of the LSI\_UNED team in the MultiClinNER sub-task for Spanish, English, and Italian using the W<sup>2</sup>NER architecture. The submitted systems followed a monolingual and single-label strategy. Overall, the results were consistent across languages, with the best performance obtained for Spanish, where data augmentation with external NER resources improved the system’s performance and obtained a third place for the three labels in strict\_f1 (including the first place in strict\_recall for the en-

tity DISEASE). English and Italian also produced competitive results, although symptom detection remained the most challenging category.

One of the main limitations of this work is that no hyperparameter optimization was performed due to time constraints. As a result, the selected configurations may not represent the optimal performance of the proposed approach.

Future work will explore more complex modeling strategies, including multi-label and multi-lingual configurations. Another direction would be to compare W<sup>2</sup>NER with alternative NER approaches, such as traditional token-level sequence labeling using IOB2 or BILOU tagging schemes, as well as more recent architectures like GLiNER (Zaratiana et al., 2024). In addition, it would be interesting to carry out hyperparameter optimization using cross-validation to identify the optimal combination of hyperparameters.

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